

4DVarNet-FM Synthesis Report

4DVarNet-FM: Experiment Synthesis Report (live-updated)

Generated: 2026-06-29 23:28:52 | T_max = 3.0s | dt = 0.01 | 15 obs/window

Part 1 – 4DVarNet-FM Variants:

Config	CS1 X	CS1 Y	CS1 Z	CS1 μ	CS2 X	CS2 Y	CS2 Z	CS2 μ	Deg	Time
B1_small_unet	0.081	0.098	0.113	0.097	0.100	0.109	0.122	0.110	1.13	x 47m
C1_longer_train	0.172	0.165	0.190	0.175	0.178	0.166	0.192	0.179	1.02	x 105m
C4_stage1_only	5.239	9.310	23.128	12.559	5.340	9.312	22.558	12.404	0.99	x 5m
D1_cs2_only	7.754	6.510	7.441	7.235	7.247	6.606	7.598	7.150	0.99	x 33m

Part 2 – DA Baselines (same test data):

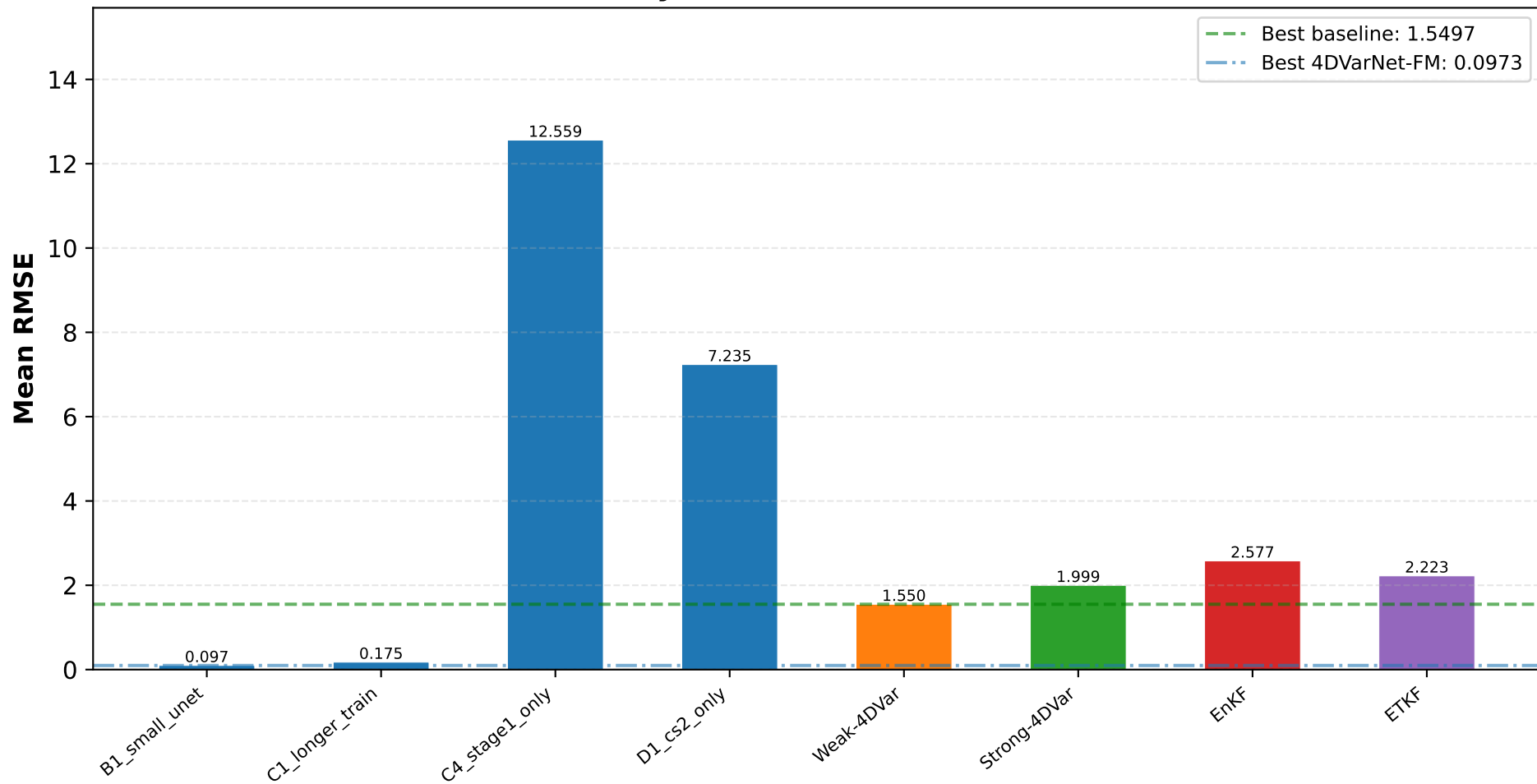
Method	CS1 μ	CS2 μ	Deg	
Weak-4DVar	0.6543	1.5497	2.37	x
Strong-4DVar	0.6604	1.9987	3.03	x
EnKF	0.7823	2.5772	3.29	x
ETKF	0.7014	2.2226	3.17	x

Best 4DVarNet-FM: B1_small_unet (CS2 μ =0.1104)

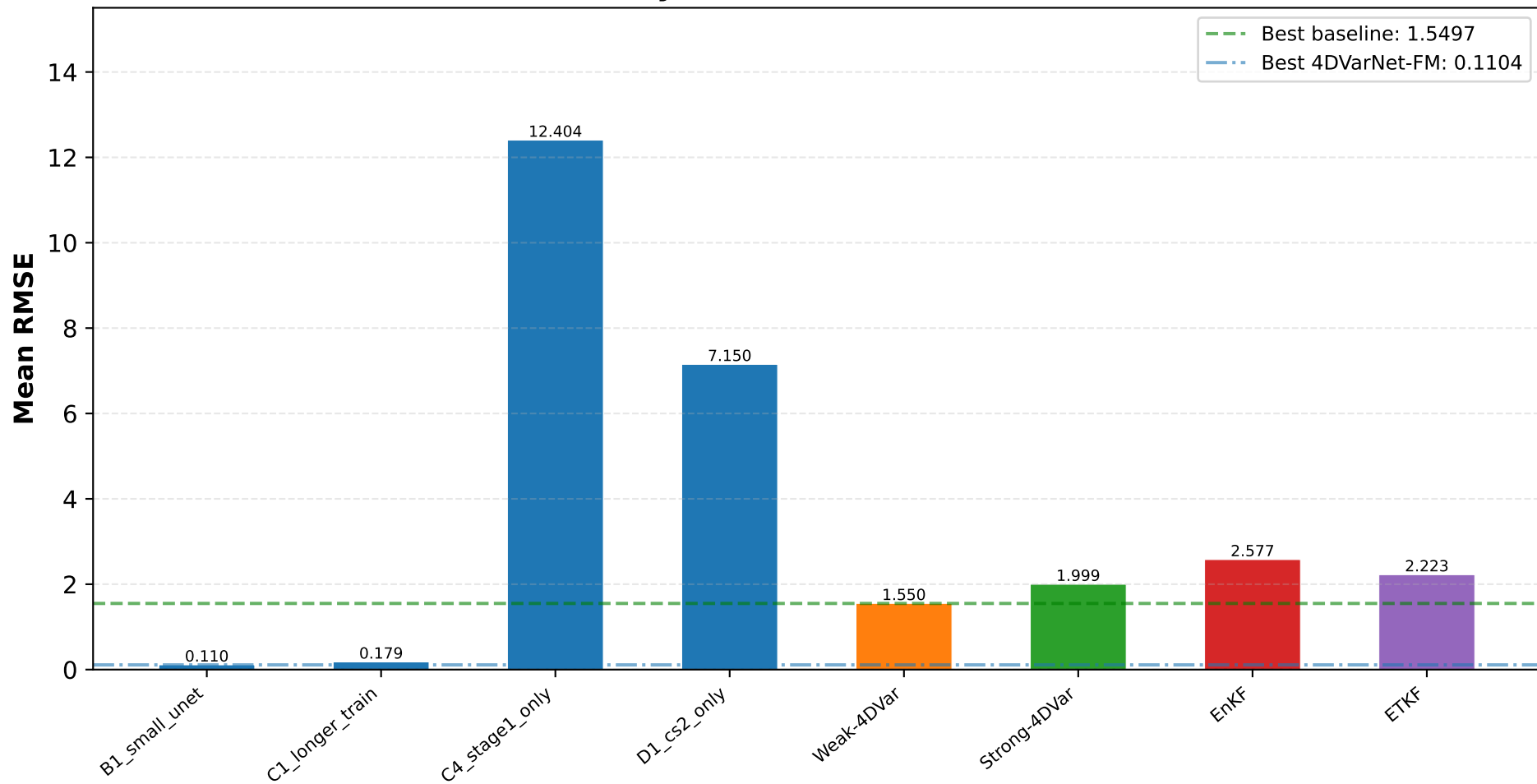
Best baseline CS1: Weak-4DVar μ =0.6543

Best baseline CS2: Weak-4DVar μ =1.5497

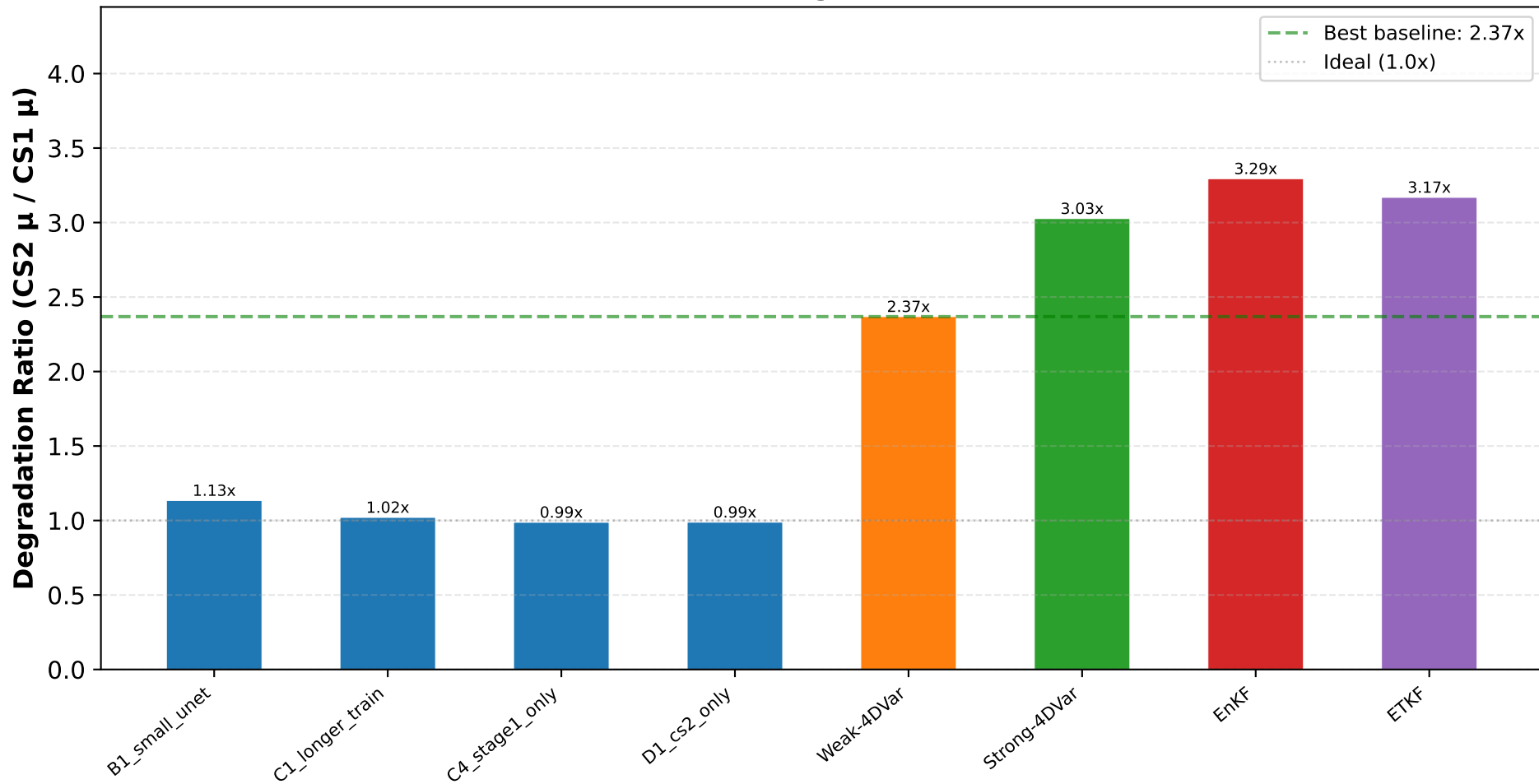
Case Study 1 — RMSE (lower is better)



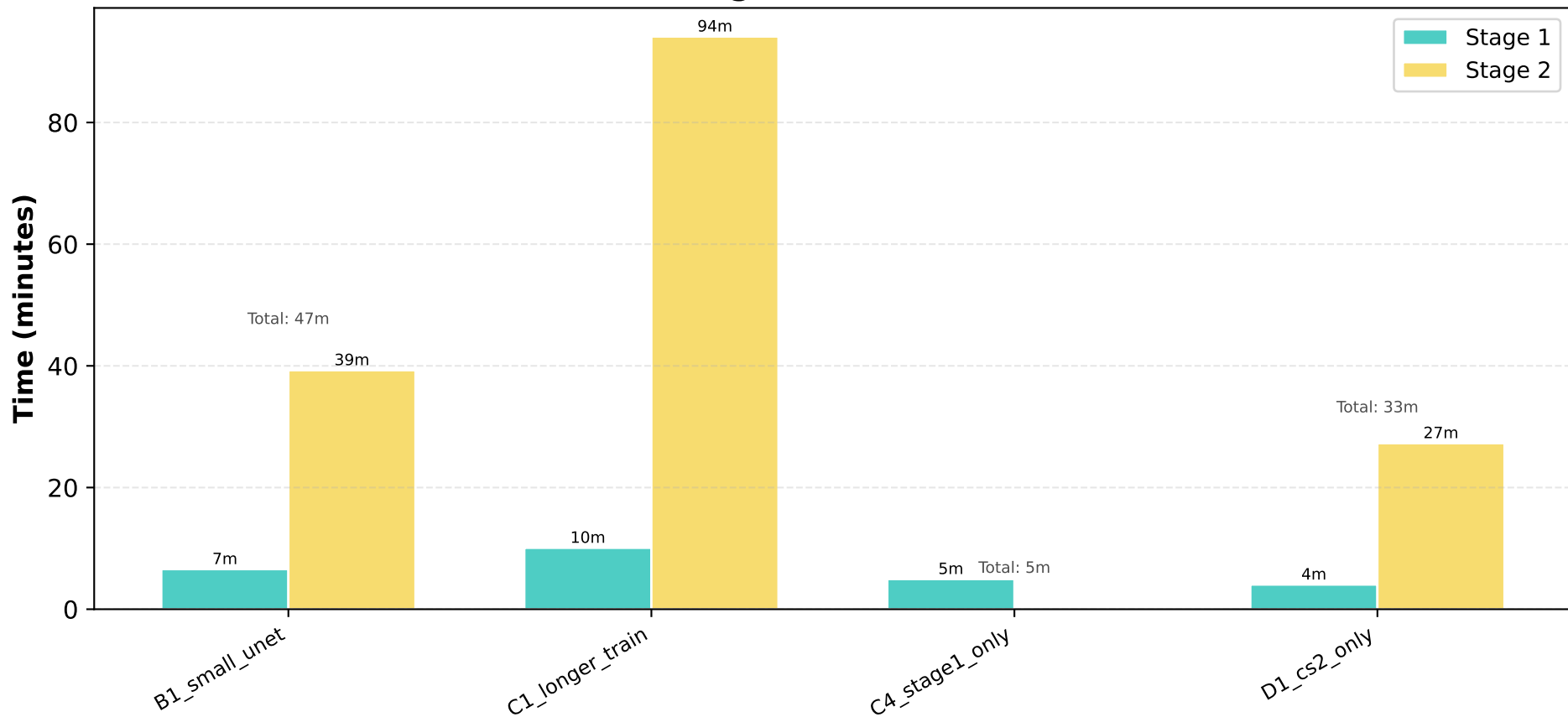
Case Study 2 — RMSE (lower is better)



Robustness — Lower Degradation = More Robust



Training Time Breakdown



4DVarNet-FM Synthesis Report — Conclusion

Conclusion & Recommendations (live-updated)
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Best 4DVarNet-FM variant: B1_small_unet
CS1 mean RMSE: 0.0973 (best baseline Weak-4DVar: 0.6543) ✓ BEATS baseline
CS2 mean RMSE: 0.1104 (best baseline Weak-4DVar: 1.5497) ✓ BEATS baseline
Degradation: 1.13x

Comparison across all variants:

B1_small_unet	CS1 $\mu=0.0973$	CS2 $\mu=0.1104$	Deg=1.13x	★ BEATS ALL BASELINES
C1_longer_train	CS1 $\mu=0.1753$	CS2 $\mu=0.1788$	Deg=1.02x	★ BEATS ALL BASELINES
C4_stage1_only	CS1 $\mu=12.5590$	CS2 $\mu=12.4035$	Deg=0.99x	Below baselines
D1_cs2_only	CS1 $\mu=7.2349$	CS2 $\mu=7.1504$	Deg=0.99x	Below baselines

Key Insights:

Recommendation:
Use B1_small_unet for production (lowest CS2 RMSE).